

Predicting Hospital Readmissions from Clinical Discharge Summaries

Alina Gavrilov¹ Kalyani Limaye¹ Nils Kuhn²

¹Institute for Computation and Mathematical Engineering, Stanford

²Department of Electrical Engineering, Stanford



Introduction

Hospital readmissions represent a significant challenge in healthcare, imposing substantial financial burdens on institutions while indicating potential gaps in patient care quality. We leverage Electronic Health Records (EHR), specifically clinical discharge summaries, as a rich data-source to feed into existing LLM's, obtain semantic embeddings, and implement several simple Machine Learning (ML) methods to predict readmission.

Data

Our study uses discharge summaries from the MIMIC-IV Note dataset, containing 331,794 summaries from 145,915 patients. Each includes admission reasons, hospital course, and discharge instructions. Binary labels created by sorting records by subject ID and chart time, setting $Y = 1$ if a future hospital visit was recorded and $Y = 0$ otherwise. Data was subset into balanced portions based on compute needs.

- **Multi-Hot Embeddings** Medically relevant keywords from Chiu et al. (2024) used to vectorize 10,000 summaries. Multi-hot embeddings capture keyword frequency per document.
- **Pre-trained ClinicalBERT** Discharge summaries split into sentences and further into 512-token chunks to prevent mid-sentence splits. Each chunk embedded using ClinicalBERT. Sentence- and document-level embeddings averaged for final representations.
- **Fine-tuned ClinicalBERT (Classification)** BioClinicalBERT fine-tuned for binary classification by adding an untrained linear layer. Model trained for 3 epochs on 179,836 samples with learning rate 5×10^{-6} , batch size 4, and 16 gradient accumulation steps.
- **Fine-tuned ClinicalBERT (Semantic Embeddings)** Trained using cosine similarity loss, penalizing embeddings with small angular distances across different labels. Used six chunks per input, batch size 2, and a cosine similarity margin 0.5, completing one epoch in 12 hrs.
- **OpenAI 4o-mini Summarization** Notes summarized using OpenAI's 4o-mini API, then processed using text-embedding-ada-002 for final representation.
- **OpenAI Full-Text Embeddings** Full-text discharge summaries directly embedded using OpenAI's text-embedding-ada-002 model.

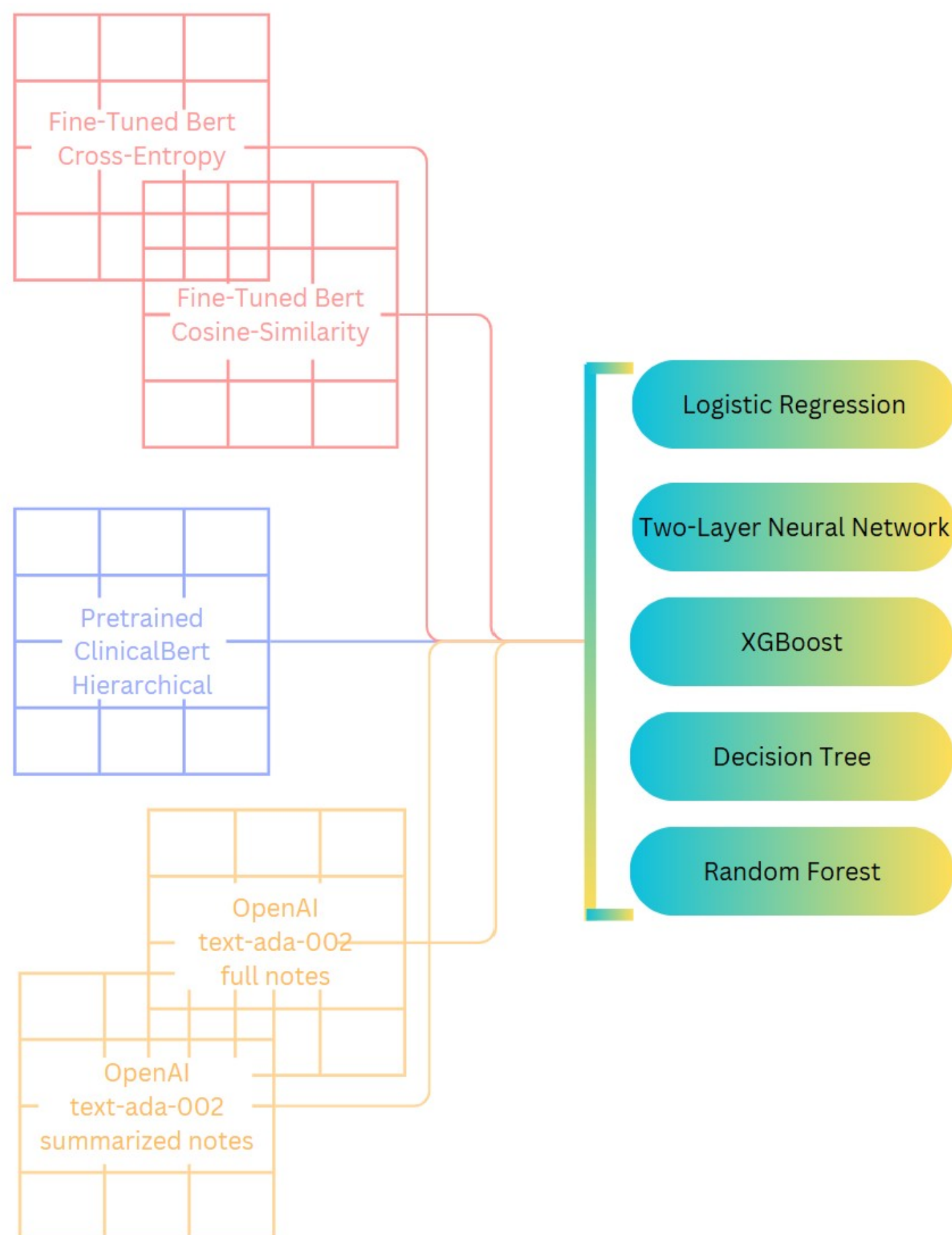


Figure 1. All Embeddings and Methods Used

Methods

The baseline model uses a multi-hot encoding embedding using keywords from existing literature [Chiu et al. (2024)] for a simple logistic regression model. This baseline achieved an accuracy of 63% on our test data.

- **Logistic Regression** A standard logistic regression model with L2 regularization ($\lambda = 1$). The bias term is not regularized. Weights were updated using gradient ascent until convergence or one million iterations. Learning rate: 0.01.
- **XGBoost (Boosting)** A boosting model that constructs up to 5000 trees with a max depth of 3. Misclassified samples are weighted more heavily in subsequent trees. We applied L2 regularization ($\lambda = 1$) and 80% sub-sampling for both rows and columns. Log loss was used as the objective function.
- **Random Forest** A bagging-based ensemble of up to 100 decision trees, each with a max depth of 8. We applied 80% sub-sampling and column sampling per tree and 80% column sampling per node. To address class imbalance, the majority class was downsampled in training and validation sets.
- **Decision Tree** A single deep decision tree without sub-sampling or regularization. This method provided the lowest overall performance due to overfitting and lack of generalization.
- **Two-Layer Neural Network** A feedforward neural network with two hidden layers of 50 neurons each, using sigmoid activations. The model used batch gradient descent with a dynamically reduced learning rate ($\alpha = 0.01$) and L2 regularization ($\lambda = 0.0025$). Evaluated using 5-fold cross-validation.

Metrics and Results

	Dataset	Fine-tuned Bert CE	Fine-tuned Bert CS	OpenAI Full Text	OpenAI Summarized	Pre-trained Bert
Model						
Decision Tree	F1:	0.63	F1: 0.58	F1: 0.61	F1: 0.61	F1: 0.66
	Acc:	0.63	Acc: 0.58	Acc: 0.52	Acc: 0.54	Acc: 0.61
Logistic Regression	F1:	0.67	F1: 0.62	F1: 0.61	F1: 0.59	F1: 0.70
	Acc:	0.67	Acc: 0.63	Acc: 0.54	Acc: 0.56	Acc: 0.65
NN	F1:	0.67	F1: 0.62	F1: 0.55	F1: 0.63	F1: 0.71
	Acc:	0.68	Acc: 0.61	Acc: 0.49	Acc: 0.55	Acc: 0.63
Random Forest	F1:	0.69	F1: 0.66	F1: 0.73	F1: 0.79	F1: 0.68
	Acc:	0.68	Acc: 0.66	Acc: 0.66	Acc: 0.74	Acc: 0.66
XGBoost	F1:	0.65	F1: 0.65	F1: 0.70	F1: 0.67	F1: 0.73
	Acc:	0.65	Acc: 0.65	Acc: 0.60	Acc: 0.60	Acc: 0.68

Figure 2. F1/Accuracy Matrix

We evaluated model performance using F1-score and accuracy to ensure a balanced assessment across both minority and majority classes. F1-score, which combines precision and recall, helps identify false positives and negatives, ensuring clinically meaningful predictions. Among the tested models, Random Forest performed best, particularly with OpenAI summaries (F1: 0.79, Accuracy: 0.74).

However, the OpenAI dataset was considerably smaller than other embedding sets, which may introduce bias.

The neural network exhibited a strong F1-score (0.71) but lower accuracy (0.63) with pre-trained ClinicalBERT, suggesting that while it captures important patterns, it may misclassify ambiguous cases.

The overall classification error remained above 30%. Part of it can be attributed to limitations such as incomplete clinical notes, hospital transfers, and unrecorded readmissions due to patient choices or mortality.

These factors highlight the challenges of using discharge summaries alone for readmission prediction.

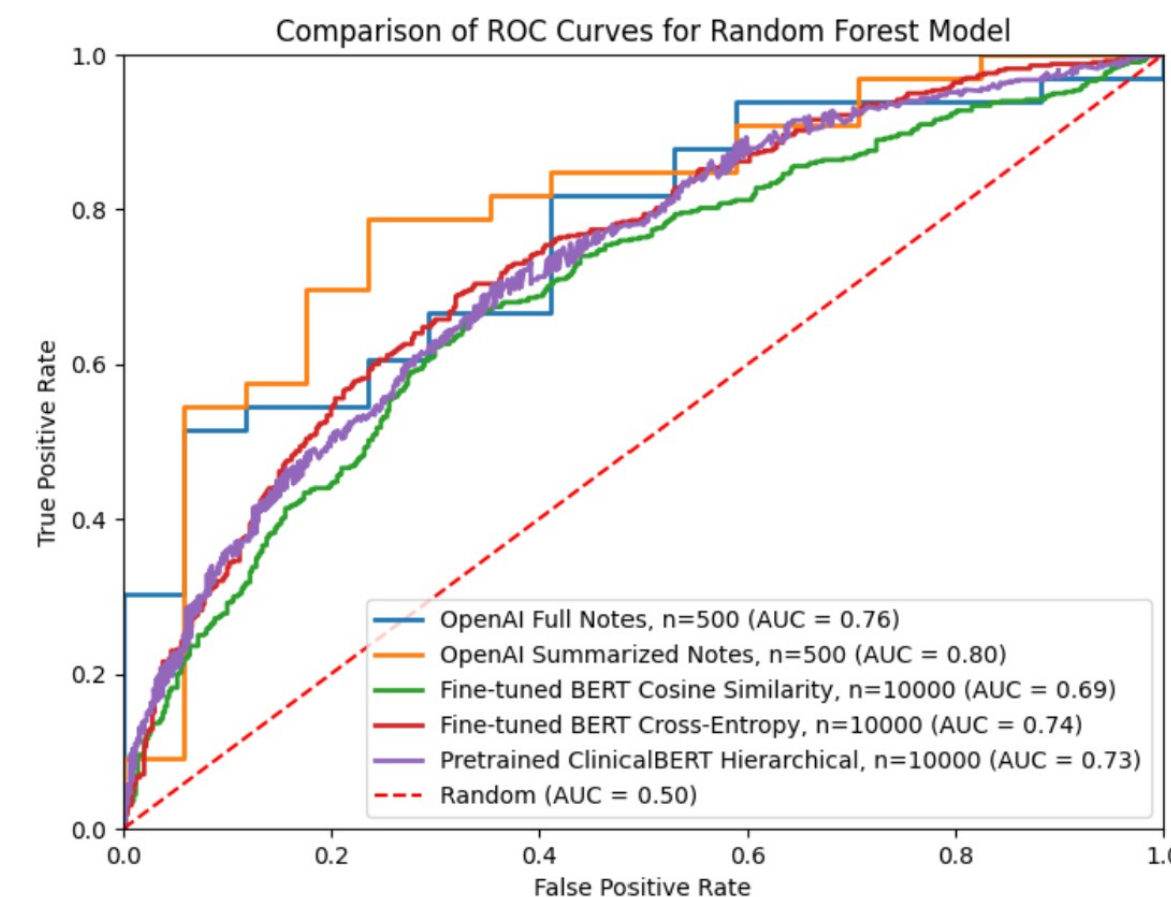


Figure 3. ROC Random Forest Models

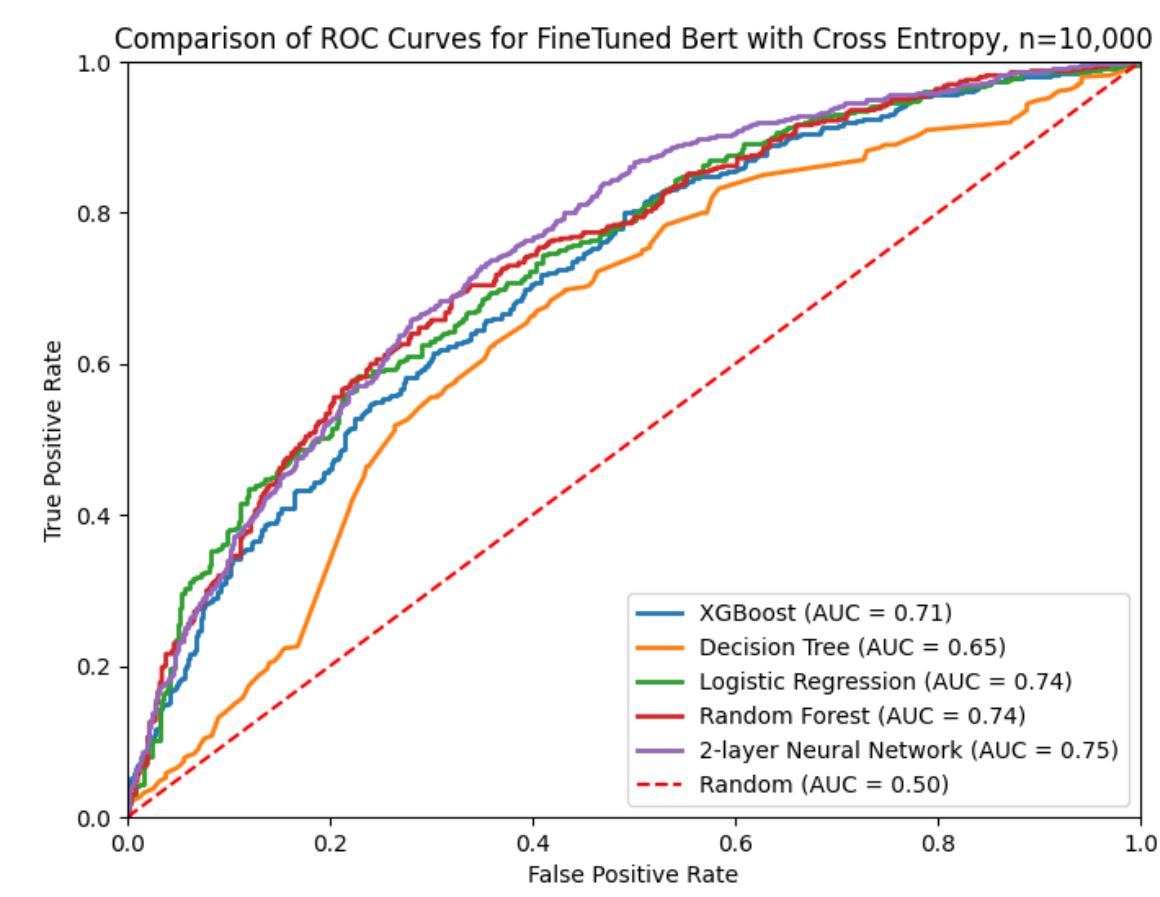


Figure 4. ROC Fine-Tuned Bert Cross-Entropy

Conclusions

The stable error rate across embedding strategies and models, suggesting meaningful signal relevant to readmission can be extracted from semantic features, yet further refinement is necessary for a more reliable predictive model. Overall, performance on the validation set and the test set differed by only about 0.02, suggesting the model does not suffer from severe overfitting. Although the error rate may point to underfitting, it is possible other factors contribute to elevated error.

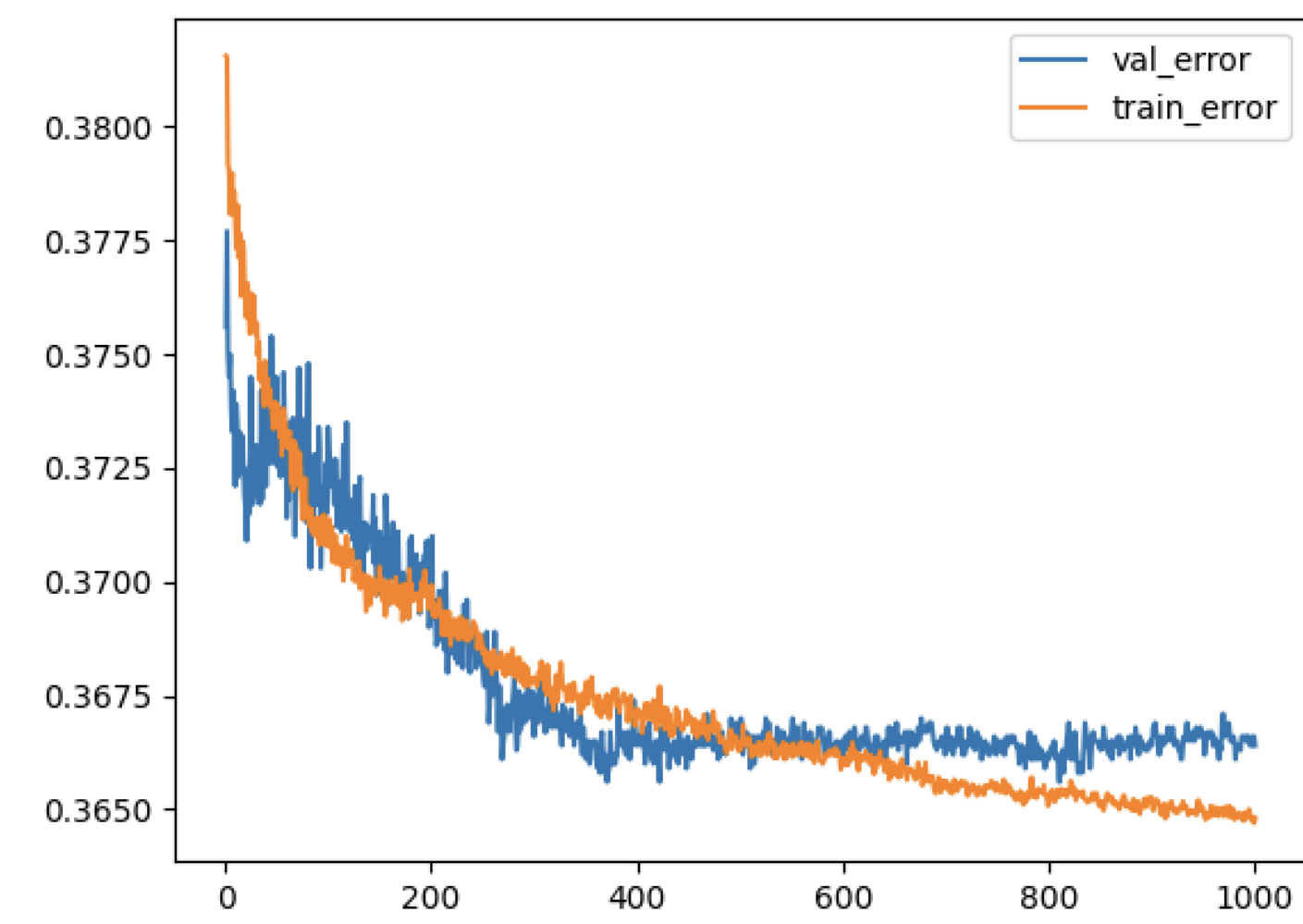


Figure 5. Neural Network Loss vs. Epochs

Future Work

With more time, we would expand our dataset to include the full 330,000+ discharge summaries, allowing for broader linguistic and clinical variability. We would fine-tune ClinicalBERT with additional epochs and explore retrieval-augmented generation (RAG) methods to enhance feature extraction. Computational efficiency would be improved by transitioning to distributed training on a high-performance computing cluster. We also plan to replace manual hyperparameter tuning with Bayesian optimization to systematically improve model performance. Additionally, we would test alternative LLMs such as BlueBERT, PubMedGPT, and MedAlpaca to determine their effectiveness for clinical readmission prediction. Given the promising results of OpenAI 4o-mini summary embeddings, we would experiment with fine-tuning this model on medical notes and explore chain-of-thought prompting to improve interpretability and predictive accuracy.

References

- [1] C.-C. Chiu, C.-M. Wu, T.-N. Chien, L.-J. Kao, and C. Li. Predicting icu readmission from electronic health records via bertopic with long short-term memory network approach. *Journal of Clinical Medicine*, 13:5503, 2024.